

Question Paper

Exam Date & Time: 29-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: ECE 2124/ECE_2124 DIGITAL SYSTEM DESIGN
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Design a 4-bit Excess-3 to BCD code converter and implement the minimized Boolean expression using NAND gates only. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Implement the logic functions $F_1(A, B, C) = \sum m(1, 4, 6)$, $F_2(A, B, C) = \sum m(3, 5)$, and $F_3(A, B, C) = \sum m(0, 2, 7)$ using a 1-to-8 demultiplexer and necessary additional logic gates. Assume that DEMUX has active low outputs. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Design a 4:16 line decoder with active-high enable and active-high outputs by using 2:4 line decoders and the necessary logic gates. (3)

Course Outcome: "CO1" / Cognitive Level: "Create"

- 2A) Design a synchronous counter that follows the counting sequence 0, 1, 3, 4, 2, and then repeats. Ensure that all unused states return to 0. Use positive edge-triggered T flip-flops for the design. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2B) Design an asynchronous decade counter using positive edge triggered JK flipflops. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2C) Design a single-digit BCD adder using two 4-bit binary adders and realize the BCD correction circuit using NAND gates. Explain the operation of the circuit with suitable examples. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3A) Design a sequential circuit using Mealy machine that can detect the non-overlapping sequence 0110. Use positive edge triggered D flipflops. (4)

Course Outcome: "CO2" / Cognitive Level: "Create"

- 3B) Apply the concept of characteristic table to convert a JK flip-flop into an equivalent D flip-flop. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

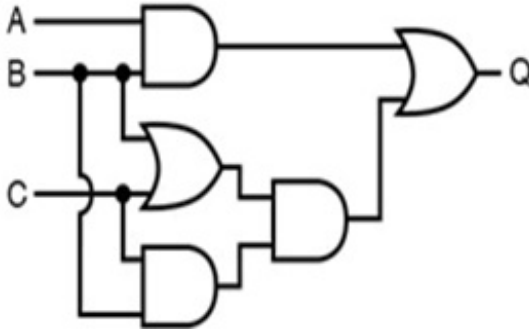
- 3C) Write a structural Verilog module for a full subtractor by hierarchically integrating half subtractors and the necessary logic gates. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Write a behavioural Verilog coding for 8:3 priority encoder using nested if statement. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4B) For the given circuit in figure 4B, write a Verilog dataflow modelling by considering a delay of 10ns for the execution of function, also draw the timing diagram for inputs and outputs. (3)



Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) Write a Verilog user defined primitive (UDP) for carryout output of full adder circuit. (3)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5A) a) Implement SR latch using ACTEL ACT-1 module.
b) Explain the architecture of ACT-2 C logic module. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5B) Explain ASIC design flow with neat flow diagram. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5C) Explain with example
i. Noise margin
ii. fan-in in digital circuit
iii. propagation delay (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_2124:DIGITAL SYSTEM DESIGN

#	Course Outcome	Description	Marks
1	1	CO 1	23
2	2	CO 2	4
3	3	CO 3	13
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	13	43(86%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	2	7(14%)
	Total	15	50 (100%)